Announcement

Since Monday the 5th of May is a bank Holiday, next weeks lectures are scheduled as follows.

- Tuesday 6th of May, 16:00-17:00 Conference Auditorium 1
- Wednesday 7th of May, 9:00-10:00 RBLT
- Tutorials are also moved to Tuesday or Thursday $\longrightarrow$ please consult your timetable for time and location!

Formal Language & Finite Automata

Undecidability

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Reading

All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 206 - 2014 (essential)

Recap

Recognizing vs. deciding

Given a Turing machine M and an input string w there are three outcomes.

1. M accepts w;
2. M rejects w; or
3. M never halts (enters an infinite loop).

Definition
A language L is **recursive enumerable** or **Turing recognizable** if it is recognized by some Turing machine, that is there is a Turing machine M such that $L(M) = L$.

Definition
A language L is **recursive** or **Turing decidable** if it is decided by some Turing machine, that is there is a Turing machine M such that $L(M) = L$ and M halts on every input string.

Existence of unrecognizable languages

Existence of unrecognizable languages

We proved that there are languages which are not Turing recognizable by a counting argument.

Theorem
There exists a language which is not Turing recognizable.

An undecidable and an unrecognizable language

An undecidable and an unrecognizable language

We define the following language:

$$A_{\mathsf{TM}} = \{\langle A, w\rangle \mid A \text{ is a TM that accepts } w\}.$$

Theorem
A_{TM} is not Turing decidable.

Theorem
$\overline{A_{\mathsf{TM}}}$ is not Turing recognizable.